Funzioni

funzione monotona $\forall x_1, x_2 \in I$

$x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$

funzioni composte:

$f(x) = g(f(x))$

$x \in \text{dom } g \circ f \iff x \in \text{dom } f \ e \ f(x) \in \text{dom } g$

$\cos^2 x + \sin^2 x = 1 \quad \forall x \in \mathbb{R}$

$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$

$\cos(\alpha \pm \beta) = \cos\alpha\sin\beta \mp \sin\alpha\cos\beta$

$\sin 2x = 2\sin x\cos x \quad , \quad \cos 2x = 2\cos^2 x - 1$

$\sin x - \sin y = 2\sin\frac{x-y}{2}\cos\frac{x+y}{2}$

$\cos x - \cos y = -2\sin\frac{x-y}{2}\sin\frac{x+y}{2}$

$\sin(x+\pi) = -\sin x \qquad \cos(x+\pi) = -\cos x$

$\sin\left(x \pm \frac{\pi}{2}\right) = \cos x \qquad \cos\left(x \pm \frac{\pi}{2}\right) = \mp\sin x$

$\tan x = \frac{\sin x}{\cos x} \quad , \quad \cot\text{an } x = \frac{\cos x}{\sin x}$

a) $3x+1 > 0 \qquad x \geq -\frac{1}{3}$

$x^2+x-6 > 0 \qquad x < -3 \vee x > 2$

↳ $(x-2)(x+3)$

$\to \forall x \in \mathbb{R} \setminus \{-3, 2\}$

b) $\frac{\sqrt{x^2-3x-4}}{x+3} \qquad x \neq -3$

↳ $(x-4)(x+1) \geq 0$

$x \geq 4 \quad x \leq -1$

$\begin{cases} \forall x \in \mathbb{R} \setminus -1 \leq x \leq 4 \\ (-\infty; -3); (-3; -1] \cup \\ \cup [4; +\infty) \end{cases}$

c) $\log(x^2 - x)$

↳ $x(x-1)$

$x < 0$

$x > 1$

$(-\infty; 0) \cup (1; +\infty)$

d) $\begin{cases} \frac{1}{2x+1} & x \geq 0 \\ e^{\sqrt{x+1}} & x < 0 \end{cases}$

$x \geq -1$

C.E.: $x > -1$

$[-1; +\infty)$

a) $f(x) = \frac{1}{x^2+1}$

Immagine $(0; 1]$

$y = f(x)$

$y = \frac{1}{x^2+1}$

$\frac{1}{y} = x^2 + 1$

$x = \sqrt{\frac{1}{y} - 1}$

↳ $\frac{1}{y} - 1 \geq 0$

$\frac{1}{y} \geq 1$

$(0; 1]$

b) $f(x) = \sqrt{x+2} - 1$

$\sqrt{\cos x - 1}$ → $\cos x - 1 \geq 0$

$\cos x \geq 1$

↳ $x = 2k\pi$

$x = \cos^{-1}(y^2 + 1)$

$\lim_{x \to +\infty} x^n = +\infty \qquad \lim_{x \to 0^+} x^n = 0$

$a > 0: \lim_{x \to +\infty} x^a = 0 \qquad \lim_{x \to 0^+} x^a = +\infty$

$\lim_{x \to +\infty} a^x = +\infty \qquad \lim_{x \to -\infty} a^x = 0 \qquad a > 1 \quad \lim_{x \to +\infty} \log_a x = +\infty \quad \lim_{x \to 0^+} \log_a x = -\infty$

$\lim_{x \to +\infty} a^x \qquad \lim_{x \to -\infty} a^x = +\infty \qquad a < 1 \quad \lim_{x \to +\infty} \log_a x = -\infty \quad \lim_{x \to 0^+} \log_a x = +\infty$

$\lim_{x \to \pm\infty} \sin x / \cos / \tan x$ non esistono

$\lim_{x \to (\frac{\pi}{2} + k\pi)} \tan x = \mp\infty \quad \forall k \in \mathbb{Z} \qquad \lim_{x \to \pm 1} \arcsin x = \pm\frac{\pi}{2} = \arcsin(\pm 1)$

$\lim_{x \to +1} \arccos x = 0 = \arccos 1 \quad , \quad \lim_{x \to -1} \arccos x = \pi = \arccos(-1)$

$\lim_{x \to \pm\infty} \arctan x = \pm\frac{\pi}{2}$

$\lim_{x \to 0} \frac{\log(1+x)}{x} = 1 \qquad \lim_{x \to 0} \frac{e^x - 1}{x} = 1$

$\lim_{x \to 0} \frac{\sin x}{x} = 1 \qquad \lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \qquad \lim_{x \to \pm\infty} \left(1 + \frac{a}{x}\right)^x = e^a$

$\lim_{x \to 0} (1+x)^{1/x} = e \qquad \lim_{x \to 0} \frac{\log_a(1+x)}{x} = \frac{1}{\log a}$

$\lim_{x \to 0} \frac{(1+x)^\alpha - 1}{x} = \alpha \qquad (\alpha \in \mathbb{R})$

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$\lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e$

$\lim (2x^2 + 3)$

$\frac{x}{\sqrt{x^2-1}} \cdot \frac{\sqrt{x^2+1}}{\sqrt{x^2+1}} = \frac{x \cdot \sqrt{x^2+1}}{x^2 - 1} =$

$\lim_{x \to -\infty} \frac{x}{\sqrt{x^2-1}} = -1$

$= \frac{x \cdot \sqrt{x^2+1}}{x^2-1} = \frac{x}{x^2+1} \cdot \frac{\sqrt{x^2+1}}{x^2-1}$